

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (CE/IT) Examination****December 2015****IT405 Advanced Computing****Date: 18.12.2015, Friday****Time: 10:00 a.m. To 1:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the below given multiple choice questions: [07]**

- a) Which of the following service is used for setting password less communication between nodes in a cluster?
(a) NFS (b) MPICH (c) SSH (d) All of the above
- b) SETI@home(Search for Extraterrestrial Intelligence)project is an example of
(a) Cluster Computing (b) Grid Computing
(c) Cloud Computing (d) Hybrid Computing
- c) Clusters are homogeneous whereas Grids are _____.
(a) Heterogeneous (b) Homogeneous (c) Distributed (d) None of the above.
- d) Two organizations connected through a virtual private network for global sharing of resources is an example of _____.
(a) Intergrid (b) Extragrid (c) Intragrid (d) Both a & b.
- e) _____ describes a distribution model in which applications are hosted by a service provider and made available to users.
(a) Infrastructure-as-a-Service (IaaS) (b) Platform-as-a-Service (PaaS)
(c) Software-as-a-Service (SaaS) (d) Cloud service
- f) Which of the following cloud concept is related to pooling and sharing of resources ?
(a) Polymorphism (b) Abstraction (c) Virtualization (d) None of the mentioned
- g) You can't count on a cloud provider maintaining your _____ in the face of government actions.
(a) scalability (b) reliability (c) privacy (d) None of the mentioned

Q – 2.a How can we measure scalability of a cluster? [04]**Q – 2.b Why System monitoring is required in Cluster Computing? Which factors should be monitored? [07]****OR****Q – 2.b Draw Cluster Computing Architecture and explain its components in detail. [07]****Q – 2.c What is Access grid? List any two security issues in access grid. [03]****OR****Q – 2.c Why Grid computing is considered as Next generation internet? [03]**

- Q - 3 Write Short Note On:(Any two)** [14]
- a. Flexible Load Sharing Algorithm in Cluster.
 - b. Creating a Virtual Organization using Grid Computing.
 - c. First Generation Grid Projects.

SECTION – II

- Q - 4 Do as directed:**
- a. State True or False: A Google doc is an example of Cloud Computing. [01]
 - b. Define Network RAM. [01]
 - c. What do you mean by Single system image in distributed computing? [01]
 - d. Give functionalities of a parallel debugger. [02]
- Q – 5.a** What are the building blocks of grid computing? [04]
- Q – 5.b** List and explain Essential characteristics of Cloud Computing. [05]

OR

- Q – 5.b** Describe the deployment models of cloud computing. [05]
- Q – 5.c** What is Infrastructure as a service? How multiple processors have made it efficient? [07]
Explain how Amazon provides solution for infrastructure as a service.

OR

- Q – 5.c** Explain Para virtualization and Full virtualization with example. [07]
- Q – 6 Attempt any Two:** [14]
- a. "Cloud computing is not always secure". Justify the statement. List the application of cloud computing in field of engineering.
 - b. Draw and explain the architecture of Xen Hypervisor. Why Domain 0 and Domain U both needed in Xen architecture?
 - c. How Cross Site trust management is handled in grid computing?
